Topology-Aware Immunization of the European Air-Travel Network: Compartmental Models and the Degree–Betweenness Question

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ABSTRACT

The topology of a mobility network decides how an epidemic spreads and which intervention contains it. We model disease propagation on the European subgraph of the OpenFlights air-travel network as a metapopulation reaction–diffusion process, running four compartmental dynamics (SIR, SIS, SEIR, SQIR) with parameters anchored to literature reproduction numbers, and we compare vaccination by random, degree, and betweenness targeting. Our central, testable question is where Europe sits on a known spectrum: in the US domestic network the largest hubs are also the main bridges, so degree- and betweenness-targeted immunization coincide, whereas in the worldwide network geopolitical community structure produces anomalous high-betweenness, low-degree gateways that only betweenness targeting captures. A subgraph- and core-aware layer (k-cores, motifs, graphlets) refines targeting and explains outcomes. The study is a reproducible pipeline whose region and transport layers (air, land, water) are parameters, so the same analysis extends across continents and to multimodal combinations. Claims are comparative, not predictive.

1. INTRODUCTION

A single immunization policy cannot be optimal for every pathogen. We ask how vaccination strategy interacts with both the infection model and the topology of the travel network, on a real, strongly heterogeneous substrate. Concretely, we simulate spread along European aviation routes and test random, degree-, and betweenness-targeted vaccination across four compartmental models, reinforcing the principle that a strategy should be matched to the available infection information rather than applied uniformly. Modelling epidemics on flight networks is not itself new; our specific positioning and the gap we fill are stated in Section 3.

2. RELATED WORK

Compartmental epidemics originate with Kermack and McKendrick [12]; network epidemiology asks how contact structure changes those dynamics [19; 11]. Scale-free topology can destroy the epidemic threshold [20], and Newman linked SIR to bond percolation for exact outbreak results [17]. Because hubs dominate spreading, targeted immunization far outperforms random [21; 3], though the best spreaders sit in the network core rather than simply at the highest degree [13]. At the inter-city scale, the air network governs global spread [4; 1], formalised as a metapopulation reaction–diffusion process with a global invasion threshold [5; 6]. Structurally, motifs [16] and graphlet signatures [22] fingerprint local topology but are rarely used to steer immunization — the gap we target.

3. NOVELTY AND POSITIONING

(1) Europe, four models, one protocol. Prior work targets the worldwide or US networks; a Europe-focused comparison across SIR, SIS, SEIR and SQIR under a common protocol is, to our knowledge, missing.
(2) The degree–betweenness question. In hub-and-bridge networks such as the US domestic system, degree and betweenness are highly correlated and the two immunization strategies give nearly identical results [25]. On the worldwide network, Guimerà et al. [8] found anomalous centrality: community and geopolitical structure give some low-degree airports very high betweenness (Anchorage being canonical). Where a network sits on this spectrum decides whether the strategies converge or diverge. We test empirically where Europe lies. Because region is a pipeline parameter (Section 7), we place Europe on the spectrum directly by repeating the analysis across continents and comparing the degree–betweenness correlation.
(3) Subgraph- and core-aware layer. We use k-core decomposition and small-subgraph statistics both to target immunization and to explain why a model–strategy pair behaves as it does, connecting the result to the local structure that produces it.

4. DATA AND NETWORK

The primary substrate is the air layer, from the OpenFlights route database [18], built with NetworkX [9]: nodes are airports, edges are routes weighted by flight frequency. The region is a parameter of network generation (Section 7), so the European subgraph is the focus while other continents and the world follow for the cross-region comparison.
Multimodal data sources. The same node set admits ground and sea layers. For land, rail and road topology is globally available from OpenStreetMap and the GRIP roads database [15]. To keep the cross-region comparison valid, flow weights must be built the same way for every region; we therefore generate them with a single gravity/radiation model calibrated on population [24], applied uniformly, and

Table 1: Per-model parameters (to be filled with cited values).

	SIR	SIS	SEIR	SQIR
R_0 (literature)	—	—	—	—
Recovery γ	—	—	—	—
Incubation σ	—	—	—	—
Quarantine κ	—	—	—	—

use Eurostat commuting data only to validate that model within Europe — never as the production source for one region while others differ. Commuting dominates short-range flux and the proxy choice materially affects predictions [1; 26]. For water, we use OpenStreetMap ferry routes and open cargo port-call data, whose network structure and disease relevance are documented in [10; 23; 14]. Air is in use; land and water enter as additional tagged layers.

Modelling assumptions. (i) Population scales with degree,

$$\mathrm{pop}(v) = P_0 + d(v)\, P_{\mathrm{route}}, \tag{1}$$

with $P_0 = 150{,}000$ and $P_{\mathrm{route}} = 45{,}000$, assuming more routes imply a larger city — a proxy, not census ground truth. (ii) Interventions are biological shields only: vaccinated cities become immune dead-ends and no nodes or edges are removed, isolating the effect of vaccination from route closures. (iii) The travel rate τ is scaled by route frequency, so busier corridors carry proportionally more travellers. (iv) The outbreak starts from a single random patient-zero seed, averaged over seeds. (v) The network is static (no timetables or seasonality). The resulting European grid is a hybridised scale-free network with extreme degree heterogeneity, in which hubs (London, Paris, Frankfurt, Amsterdam) connect high-volume corridors.

5. MODELS AND PARAMETERS

5.1 Metapopulation reaction–diffusion

Each airport carries a full compartmental model and individuals diffuse along edges [5]. Per day, a reaction step applies the force of infection $\lambda_i = \beta\, I_i/N_i$ with model-specific transitions at rates γ (recovery), σ (incubation, SEIR) and κ (quarantine, SQIR); a diffusion step moves a fraction of each compartment along outbound routes, split by route weight w_{ij} and a global travel rate τ. The four models span permanent immunity (SIR: measles), no immunity (SIS: common cold), latency (SEIR: COVID-19, SARS) and isolation (SQIR: Ebola). This formulation carries a global invasion threshold below which spread stays local [6], and is substrate-agnostic, so it applies unchanged to land and water layers.

5.2 Parameter choice

We fix clinical rates (γ, σ, κ) from each disease's epidemiology and set β to a literature R_0. On a heterogeneous network the well-mixed identity $R_0 = \beta/\gamma$ fails: degree fluctuations inflate it as $R_0 \propto \frac{\beta}{\gamma}\frac{\langle k^2\rangle}{\langle k\rangle}$ [20], and for multi-compartment models we compute R_0 via the next-generation matrix [27].

5.3 Calibration vs. verification

We do not calibrate R_0 to case data as predictive frameworks do [2], because the OpenFlights substrate carries no ground-truth outbreak to fit. Instead we (i) select an operating point in the R_0 regime where strategies actually differ, relative to the threshold [6]; (ii) run a sensitivity sweep showing the ranking of strategies is stable across β, τ, coverage and efficacy; and (iii) verify the implementation against analytic limits (single-node SIR; bond-percolation final size [17]). This recovers calibration's credibility without fitting, and keeps our claims comparative rather than predictive of absolute case counts.

6. VACCINATION STRATEGIES

With a fixed budget of 15 cities at 75% coverage and 85% efficacy, we compare: control (none); random; degree (densest traffic); and betweenness (structural bridges). We also contrast low (1%) versus high (75%) coverage. As a structural refinement we add a subgraph/core strategy: prioritise innermost k-shells [13] and cities embedded in dense transmission motifs [16; 22] rather than isolated high-degree nodes; the same statistics explain why an intervention dismantles the core or merely trims its periphery.

Expected mechanisms (hypotheses). The same strategy should act differently per model, which is the point of testing all four. Targeting high-betweenness hubs severs Europe's structural bridges [21], and we hypothesise: for SIR (permanent immunity) this boxes the outbreak into its origin region, where it starves as local susceptibles deplete — a firebreak; for SIS (no immunity), which cannot be eradicated and reaches an endemic plateau, hub immunization acts instead as a continuous filter that compresses the active baseline; for SEIR (latency), where pre-symptomatic passengers defeat departure screening, pre-vaccinating hubs intercepts them at layover nodes; and for SQIR (isolation), shifting identified hub cases into quarantine cuts the busiest corridors. These are hypotheses to be tested, not results.

7. PIPELINE AND SUBSTRATE

The study is a reproducible three-stage pipeline — data retrieval, network generation, evaluation — in which a run is a pure function of its configuration and seed, and the sweep (regions × layer combinations × models × strategies × coverage × seeds) is orchestrated with Nextflow [7]. Because the reaction step is substrate-agnostic, network generation emits every combination of the air, land and water layers of Section 4, making multimodality a comparable axis. Layers are kept tagged with their own travel rates, since coupling can cross the epidemic threshold even when no single layer would [6].

8. PLANNED EVALUATION

For each model, strategy and region we will report the network characterisation, the compartment time series over the horizon, and the total active infected population across strategies, delivered both as static figures (Gephi) and as interactive, browser-based visualisations (network views and a geo-animated outbreak). The decisive structural measurement is the rank correlation $\rho(\text{degree}, \text{betweenness})$ and any anomalous gateways: high correlation places Europe in the US-like regime (strategies coincide); low correlation, or salient anomalous nodes, places it in the worldwide-like regime (strategies diverge), which is the sharp question of Section 3.

9. CONCLUSIONS

We frame a Europe-specific, multi-model test of whether immunization strategy must be matched to network topology, centred on the degree–betweenness question and supported by a subgraph-aware layer and a reproducible, region- and layer-parametric pipeline. Results, and one explicit limitation, will follow.

10. ADDITIONAL AUTHORS

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11. REFERENCES

[1] D. Balcan, V. Colizza, B. Gonçalves, H. Hu, J. J. Ramasco, and A. Vespignani. Multiscale mobility networks and the spatial spreading of infectious diseases. Proceedings of the National Academy of Sciences, 106(51):21484–21489, 2009.

[2] D. Balcan, B. Gonçalves, H. Hu, J. J. Ramasco, V. Colizza, and A. Vespignani. Modeling the spatial spread of infectious diseases: The GLobal epidemic and mobility computational model. Journal of Computational Science, 1(3):132–145, 2010.

[3] R. Cohen, S. Havlin, and D. ben Avraham. Efficient immunization strategies for computer networks and populations. Physical Review Letters, 91(24):247901, 2003.

[4] V. Colizza, A. Barrat, M. Barthélemy, and A. Vespignani. The role of the airline transportation network in the prediction and predictability of global epidemics. Proceedings of the National Academy of Sciences, 103(7):2015–2020, 2006.

[5] V. Colizza, R. Pastor-Satorras, and A. Vespignani. Reaction–diffusion processes and metapopulation models in heterogeneous networks. Nature Physics, 3(4):276–282, 2007.

[6] V. Colizza and A. Vespignani. Invasion threshold in heterogeneous metapopulation networks. Physical Review Letters, 99(14):148701, 2007.

[7] P. Di Tommaso, M. Chatzou, E. W. Floden, P. P. Barja, E. Palumbo, and C. Notredame. Nextflow enables reproducible computational workflows. Nature Biotechnology, 35(4):316–319, 2017.

[8] R. Guimerà, S. Mossa, A. Turtschi, and L. A. N. Amaral. The worldwide air transportation network: Anomalous centrality, community structure, and cities' global roles. Proceedings of the National Academy of Sciences, 102(22):7794–7799, 2005.

[9] A. A. Hagberg, D. A. Schult, and P. J. Swart. Exploring network structure, dynamics, and function using NetworkX. In G. Varoquaux, T. Vaught, and J. Millman, editors, Proceedings of the 7th Python in Science Conference (SciPy), pages 11–15, Pasadena, CA, USA, 2008.

[10] P. Kaluza, A. Kölzsch, M. T. Gastner, and B. Blasius. The complex network of global cargo ship movements. Journal of the Royal Society Interface, 7(48):1093–1103, 2010.

[11] M. J. Keeling and K. T. D. Eames. Networks and epidemic models. Journal of the Royal Society Interface, 2(4):295–307, 2005.

[12] W. O. Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. Proceedings of the Royal Society A, 115(772):700–721, 1927.

[13] M. Kitsak, L. K. Gallos, S. Havlin, F. Liljeros, L. Muchnik, H. E. Stanley, and H. A. Makse. Identification of influential spreaders in complex networks. Nature Physics, 6(11):888–893, 2010.

[14] L. Mari, E. Bertuzzo, L. Righetto, R. Casagrandi, M. Gatto, I. Rodriguez-Iturbe, and A. Rinaldo. Modelling cholera epidemics: The role of waterways, human mobility and sanitation. Journal of the Royal Society Interface, 9(67):376–388, 2012.

[15] J. R. Meijer, M. A. J. Huijbregts, K. C. G. J. Schotten, and A. M. Schipper. Global patterns of current and future road infrastructure. Environmental Research Letters, 13(6):064006, 2018.

[16] R. Milo, S. Shen-Orr, S. Itzkovitz, N. Kashtan, D. Chklovskii, and U. Alon. Network motifs: Simple building blocks of complex networks. Science, 298(5594):824–827, 2002.

[17] M. E. J. Newman. Spread of epidemic disease on networks. Physical Review E, 66(1):016128, 2002.

[18] OpenFlights. OpenFlights: Airport, airline and route data. https://openflights.org/data.html, 2024. Accessed June 2026.

[19] R. Pastor-Satorras, C. Castellano, P. Van Mieghem, and A. Vespignani. Epidemic processes in complex networks. Reviews of Modern Physics, 87(3):925–979, 2015.

[20] R. Pastor-Satorras and A. Vespignani. Epidemic spreading in scale-free networks. Physical Review Letters, 86(14):3200–3203, 2001.

[21] R. Pastor-Satorras and A. Vespignani. Immunization of complex networks. Physical Review E, 65(3):036104, 2002.

[22] N. Pržulj. Biological network comparison using graphlet degree distribution. Bioinformatics, 23(2):e177–e183, 2007.

[23] H. Seebens, M. T. Gastner, and B. Blasius. The risk of marine bioinvasion caused by global shipping. Ecology Letters, 16(6):782–790, 2013.

[24] F. Simini, M. C. González, A. Maritan, and A.-L. Barabási. A universal model for mobility and migration patterns. Nature, 484(7392):96–100, 2012.

[25] T. Tanaka. Todo – us airport network: degree and betweenness correlation. TODO, 2014. PLACEHOLDER – verify before submission.

[26] M. Tizzoni, P. Bajardi, A. Decuyper, G. Kon Kam King, C. M. Schneider, V. Blondel, Z. Smoreda, M. C. González, and V. Colizza. On the use of human mobility proxies for modeling epidemics. PLOS Computational Biology, 10(7):e1003716, 2014.

[27] P. van den Driessche and J. Watmough. Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. Mathematical Biosciences, 180(1–2):29–48, 2002.